(1) what are the different protocols you observe at the following layer of protocol stack?

⇒ Different protocols observed at different layers of the protocol stack

(A) Application Layer protocols

Protocol	Packet count	Percentage of total packets
TLS	2991	37.8%
HTTP	34	0.4%
DNS	158	2.0%
OCSP	25	0.3%
QUIC	467	5.9%

(B) Transport layer protocols

Protocol	Packet count	Percentage of total packet
TCP	7224	91.3%
UDP	631	8.0%

(C) Network layer protocols

Protocol	Packet Count	Percentage of total count
IPv4	7866	99.4%
IPv6	15	0.2%

Wireshark · Protocol Hierarchy Statistics · LAB_CN_02.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	7911	100.0	7286510	141 k	0	0	0	7911
Ethernet	100.0	7911	1.5	111144	2,165	0	0	0	7911
Internet Protocol Version 6	0.2	15	0.0	688	13	0	0	0	15
User Datagram Protocol	0.1	4	0.0	32	0	0	0	0	4
Link-local Multicast Name Resolution	0.1	4	0.0	176	3	4	176	3	4
Internet Control Message Protocol v6	0.1	11	0.0	308	6	11	308	6	11
Internet Protocol Version 4	99.4	7866	2.2	157364	3,066	0	0	0	7866
User Datagram Protocol	8.0	631	0.1	5048	98	0	0	0	631
QUIC IETF	5.9	467	4.9	359160	6,997	467	340404	6,632	516
Link-local Multicast Name Resolution	0.1	4	0.0	176	3	4	176	3	4
Dynamic Host Configuration Protocol	0.0	2	0.0	597	11	2	597	11	2
Domain Name System	2.0	158	0.2	12851	250	158	12851	250	158
Transmission Control Protocol	91.3	7224	3.3	239272	4,661	4145	140708	2,741	7224
Transport Layer Security	37.8	2991	75.3	5488462	106 k	2990	4978433	96 k	3048
Data	0.0	1	0.0	490	9	1	490	9	1
Hypertext Transfer Protocol	0.4	34	0.2	13410	261	6	1551	30	34
Portable Network Graphics	0.0	1	0.0	124	2	1	124	2	1
Online Certificate Status Protocol	0.3	25	0.1	6161	120	25	6161	120	25
Line-based text data	0.0	2	0.0	2536	49	2	2536	49	2
Data	0.7	54	1.4	104310	2,032	54	104310	2,032	54
Internet Group Management Protocol	0.1	11	0.0	176	3	11	176	3	11
Address Resolution Protocol	0.4	30	0.0	840	16	30	840	16	30

No display filter.

✕ Close | Copy | Protocols | Help

(2) What is the total amount of data being received for the following two case?

⇒ (1) Neverssl.com → • 2514 bytes is data sent from A→B (data sent from your computer to the server)

• 1185 bytes is data received by computer B → A

(2) Cornell.edu → Total packet = 200 out of which
A→B (80 packet) = 7kb
B→B (80 packet) = 13kb

```
HTTP/1.1 301 Moved Permanently
Date: Thu, 09 Jul 2026 06:47:09 GMT
Server: Apache/2.4.66 ()
Location: http://lushsilverclearpathway.neverssl.com/online/
Content-Length: 298
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
```

Packet 1507. 0 client pkts, 3 server pkts, 0 turns. Click to select.

34.223.124.45:80 → 192.168.43.187:38068 (2,514 bytes) Show as ASCII No delta times Stream 60

Find: Case sensitive Find Next

Filter Out This Stream Print Save as… Back × Close Help

```
1511 106.468336  192.168.43.187  34.223.124.45   TCP   66 38068 → 80 [ACK] Seq=786 Ack=2099 Win=62336 Len=0 TSval=2866319508 TSecr=3419663893
1512 106.584653  192.168.43.187  34.223.124.45   HTTP  466 GET /favicon.ico HTTP/1.1
1513 106.979391  34.223.124.45   192.168.43.187  HTTP  482 HTTP/1.1 200 OK  (PNG)
1514 107.024229  192.168.43.187  34.223.124.45   TCP   66 38068 → 80 [ACK] Seq=1186 Ack=2515 Win=61952 Len=0 TSval=2866320064 TSecr=3419664322
1519 112.000004  34.223.124.45   192.168.43.187  TCP   66 80 → 38068 [FIN, ACK] Seq=2515 Ack=1186 Win=30080 Len=0 TSval=3419669326 TSecr=2866320064
1520 112.000341  192.168.43.187  34.223.124.45   TCP   66 38068 → 80 [FIN, ACK] Seq=1186 Ack=2516 Win=61952 Len=0 TSval=2866325040 TSecr=3419669326
1521 112.304611  34.223.124.45   192.168.43.187  TCP   66 80 → 38068 [ACK] Seq=2516 Ack=1187 Win=30080 Len=0 TSval=3419669723 TSecr=2866325040
```

```
GET /online HTTP/1.1
Host: lushsilverclearpathway.neverssl.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Referer: http://neverssl.com/
Upgrade-Insecure-Requests: 1
Priority: u=0, i

GET /online/ HTTP/1.1
Host: lushsilverclearpathway.neverssl.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0
```

Packet 1505. 3 client pkts, 0 server pkts, 0 turns. Click to select.

192.168.43.187:38068 → 34.223.124.45:80 (1,185 bytes) | Show as ASCII | No delta times | Stream 60

Find: | Case sensitive | Find Next

Filter Out This Stream | Print | Save as… | Back | × Close | Help

1511	106.468336	192.168.43.187	34.223.124.45	TCP	66 38068 → 80 [ACK] Seq=786 Ack=2099 Win=62336 Len=0 TSval=2866319508 TSecr=3419663893
1512	106.584653	192.168.43.187	34.223.124.45	HTTP	466 GET /favicon.ico HTTP/1.1
1513	106.979391	34.223.124.45	192.168.43.187	HTTP	482 HTTP/1.1 200 OK (PNG)
1514	107.024229	192.168.43.187	34.223.124.45	TCP	66 38068 → 80 [ACK] Seq=1186 Ack=2515 Win=61952 Len=0 TSval=2866320064 TSecr=3419664322
1519	112.000004	34.223.124.45	192.168.43.187	TCP	66 80 → 38068 [FIN, ACK] Seq=2515 Ack=1186 Win=30080 Len=0 TSval=3419669326 TSecr=2866320064
1520	112.000341	192.168.43.187	34.223.124.45	TCP	66 38068 → 80 [FIN, ACK] Seq=1186 Ack=2516 Win=61952 Len=0 TSval=2866325040 TSecr=3419669326
1521	112.304611	34.223.124.45	192.168.43.187	TCP	66 80 → 38068 [ACK] Seq=2516 Ack=1187 Win=30080 Len=0 TSval=3419669723 TSecr=2866325040

(3) Answer the following when you access the site http://neverssl.com/.

(A) How many HTTP GET requests you observe? List down the GET requests

⇒ There are 3 HTTP GET request from neverssl.com we observe:-

① 458 GET/online HTTP/1.1

② 459 GET/online/HTTP 1.1

③ 466 GET/favicon.ico HTTP/1.1

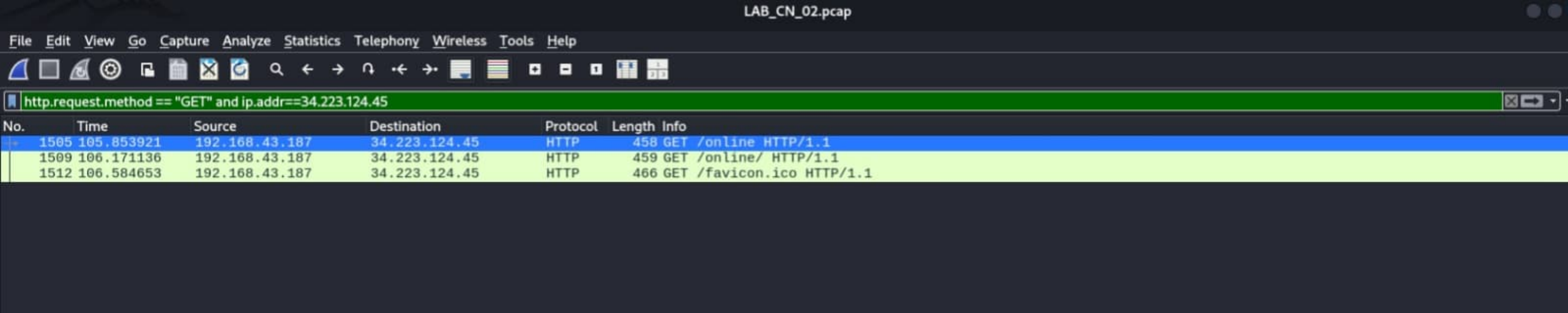
(B) What version of HTTP the server is running?

⇒ Only HTTP/1.1 version, the server is running

(C) For each of the HTTP GET requests, find out:

(i) The total number of TCP segments being received

(ii) The total amount of data being received in the corresponding HTTP response message


LAB_CN_02.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "GET" and ip.addr==34.223.124.45
No. Time Source Destination Protocol Length Info
1505 105.853921 192.168.43.187 34.223.124.45 HTTP 458 GET /online HTTP/1.1
1509 106.171136 192.168.43.187 34.223.124.45 HTTP 459 GET /online/ HTTP/1.1
1512 106.584653 192.168.43.187 34.223.124.45 HTTP 466 GET /favicon.ico HTTP/1.1

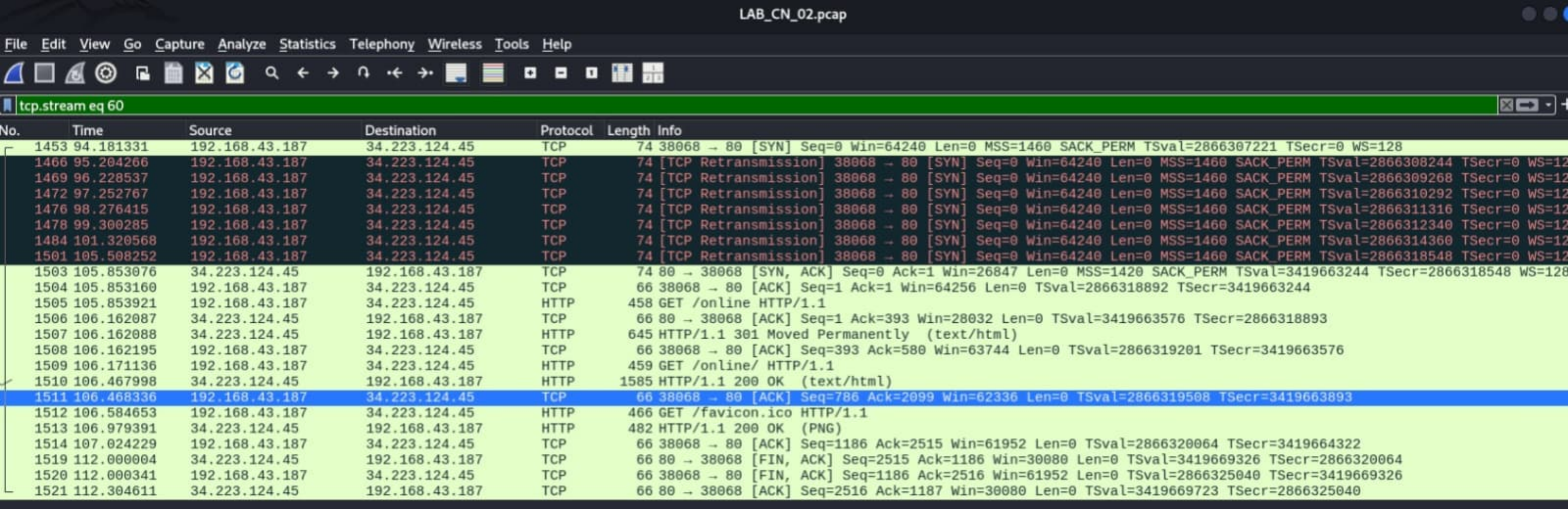
⇒

HTTP GET Request	HTTP Response	TCP Segments carrying response	Response Data
GET /online HTTP/1.1	HTTP/1.1 301 Moved permanently	1	645 bytes
GET /online /HTTP/1.1	HTTP/1.1 200 ok (text/html)	1	1585 bytes
GET /favicon.ico HTTP/1.1	HTTP/1.1 200 ok (PNG)	1	482 bytes

(D) Identify TCP three way handshake used to establish the connection with the http://neverssl.com. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three Accesses

⇒ using filter tcp.stream eq 60, the TCP three-way handshake packet are:

Step	Source	Destination	TCP Flag	Sequence No.	Acknowledgment No.
1	192.168.43.187	34.223.124.45	$\boxed{\text{SYN}}$	Seq = 0	–
2.	34.223.124.45	192.168.43.187	$\boxed{\text{SYN, ACK}}$	Seq = 0	Ack = 1
3.	192.168.43.147	34.223.124.45	$\boxed{\text{ACK}}$	Seq = 1	Ack = 1



LAB_CN_02.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.stream eq 60
No. Time Source Destination Protocol Length Info
1453 94.181331 192.168.43.187 34.223.124.45 TCP 74 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866307221 TSecr=0 WS=128
1466 95.204266 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866308244 TSecr=0 WS=12
1469 96.228537 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866309268 TSecr=0 WS=12
1472 97.252767 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866310292 TSecr=0 WS=12
1476 98.276415 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866311316 TSecr=0 WS=12
1478 99.300285 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866312340 TSecr=0 WS=12
1484 101.320568 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866314360 TSecr=0 WS=12
1501 105.508252 192.168.43.187 34.223.124.45 TCP 74 [TCP Retransmission] 38068 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2866318548 TSecr=0 WS=12
1503 105.853076 34.223.124.45 192.168.43.187 TCP 74 80 → 38068 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1420 SACK_PERM TSval=3419663244 TSecr=2866318548 WS=128
1504 105.853160 192.168.43.187 34.223.124.45 TCP 66 38068 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2866318892 TSecr=3419663244
1505 105.853921 192.168.43.187 34.223.124.45 HTTP 458 GET /online HTTP/1.1
1506 106.162087 34.223.124.45 192.168.43.187 TCP 66 80 → 38068 [ACK] Seq=1 Ack=393 Win=28032 Len=0 TSval=3419663576 TSecr=2866318893
1507 106.162088 34.223.124.45 192.168.43.187 HTTP 645 HTTP/1.1 301 Moved Permanently (text/html)
1508 106.162195 192.168.43.187 34.223.124.45 TCP 66 38068 → 80 [ACK] Seq=393 Ack=580 Win=63744 Len=0 TSval=2866319201 TSecr=3419663576
1509 106.171136 192.168.43.187 34.223.124.45 HTTP 459 GET /online/ HTTP/1.1
1510 106.467998 34.223.124.45 192.168.43.187 HTTP 1585 HTTP/1.1 200 OK (text/html)
1511 106.468336 192.168.43.187 34.223.124.45 TCP 66 38068 → 80 [ACK] Seq=786 Ack=2099 Win=62336 Len=0 TSval=2866319508 TSecr=3419663893
1512 106.584653 192.168.43.187 34.223.124.45 HTTP 466 GET /favicon.ico HTTP/1.1
1513 106.979391 34.223.124.45 192.168.43.187 HTTP 482 HTTP/1.1 200 OK (PNG)
1514 107.024229 192.168.43.187 34.223.124.45 TCP 66 38068 → 80 [ACK] Seq=1186 Ack=2515 Win=61952 Len=0 TSval=2866320064 TSecr=3419664322
1519 112.000004 34.223.124.45 192.168.43.187 TCP 66 80 → 38068 [FIN, ACK] Seq=2515 Ack=1186 Win=30080 Len=0 TSval=3419669326 TSecr=2866320064
1520 112.000341 192.168.43.187 34.223.124.45 TCP 66 38068 → 80 [FIN, ACK] Seq=1186 Ack=2516 Win=61952 Len=0 TSval=2866325040 TSecr=3419669326
1521 112.304611 34.223.124.45 192.168.43.187 TCP 66 80 → 38068 [ACK] Seq=2516 Ack=1187 Win=30080 Len=0 TSval=3419669723 TSecr=2866325040